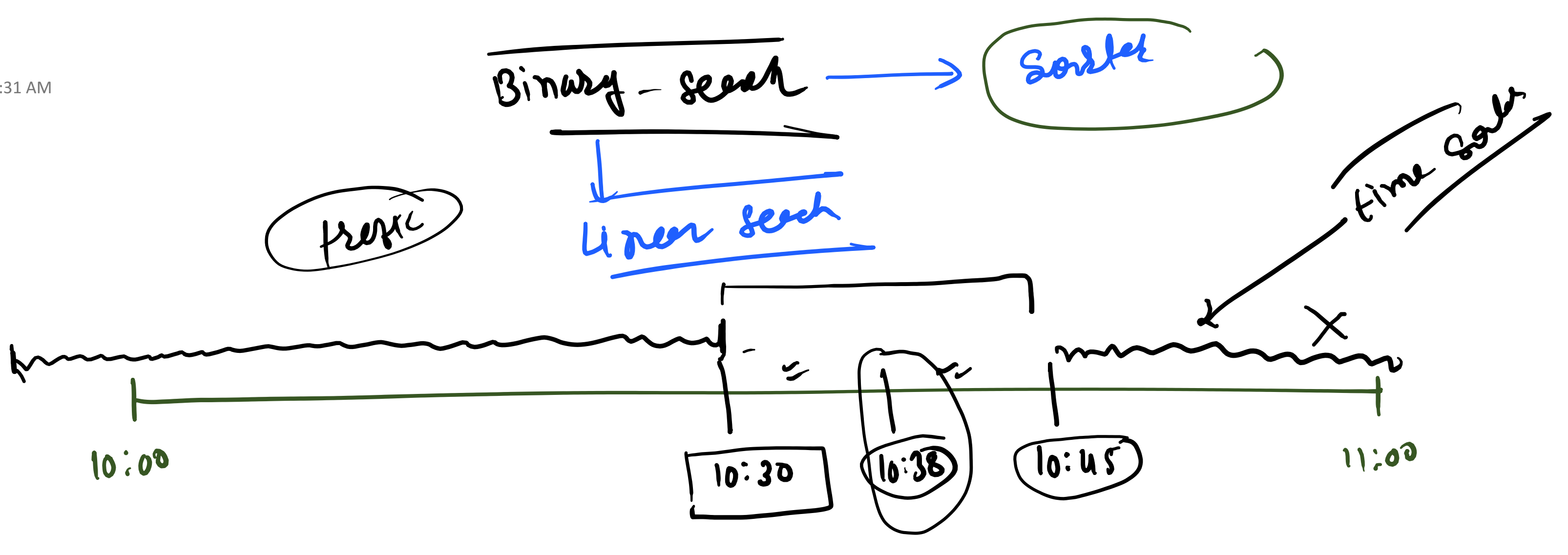
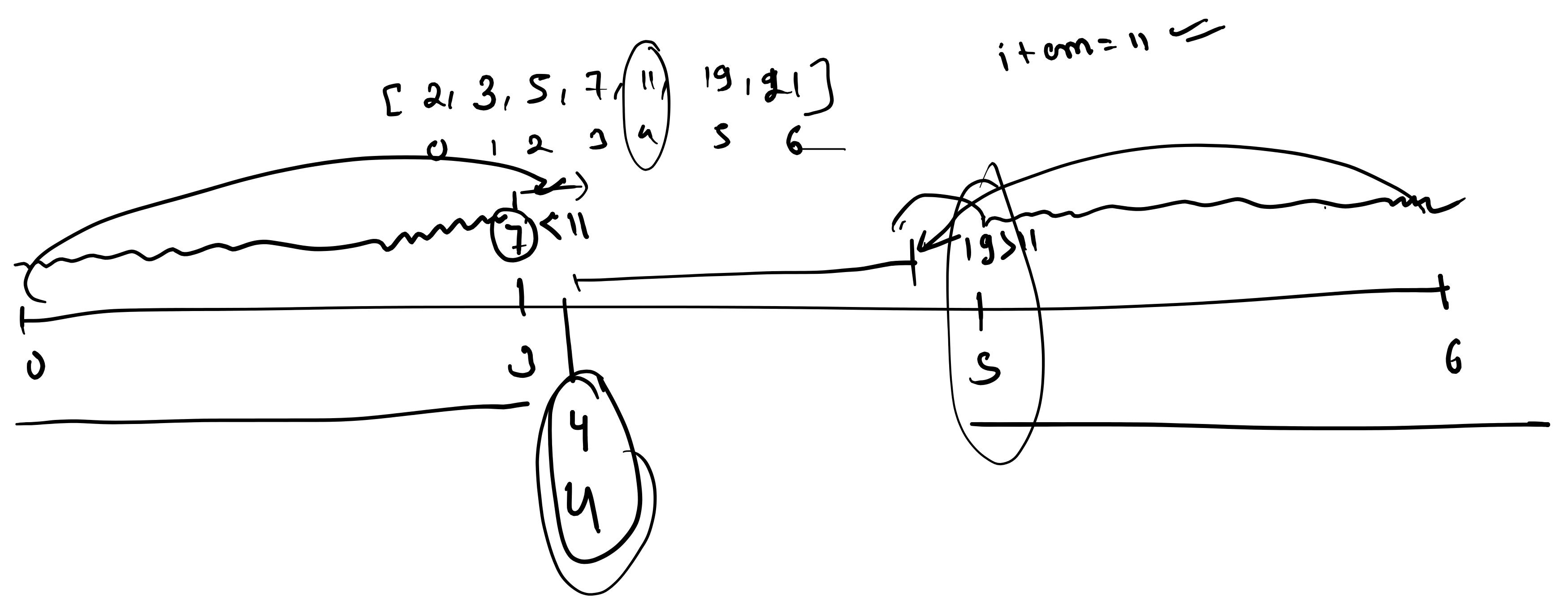
Lec-10

07/11/26 11:31 AM





$\boxed{x^K \le N}$

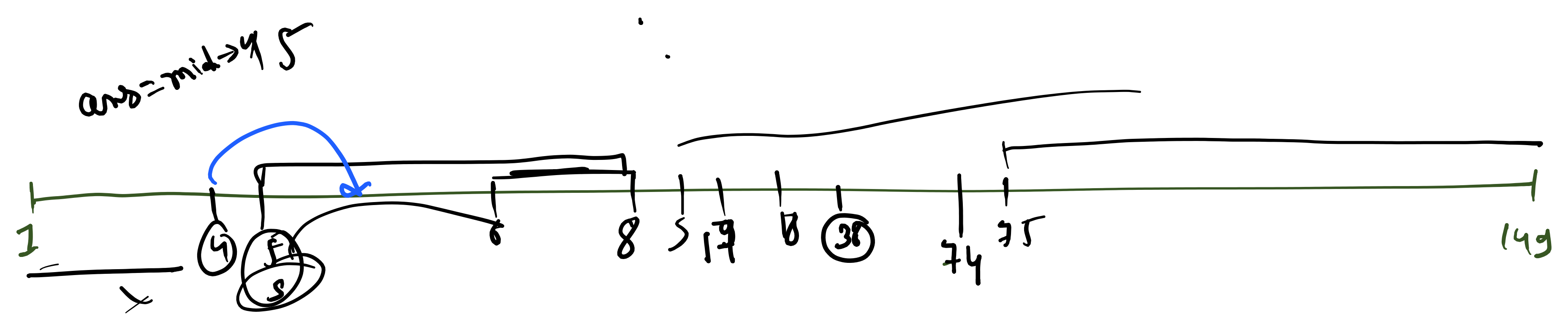
K = 3
N = 149

$n - 10^{18}$

x = 1 $1^3 \le 149$
x = 2 $2^3 \le 149$
x = 3 $3^3 \le 149$
x = 4 $4^3 \le 149$
x = 5 $5^3 \le 149$ ← 5
x = 6 $6^3 \le 149$

Linear Sea ←

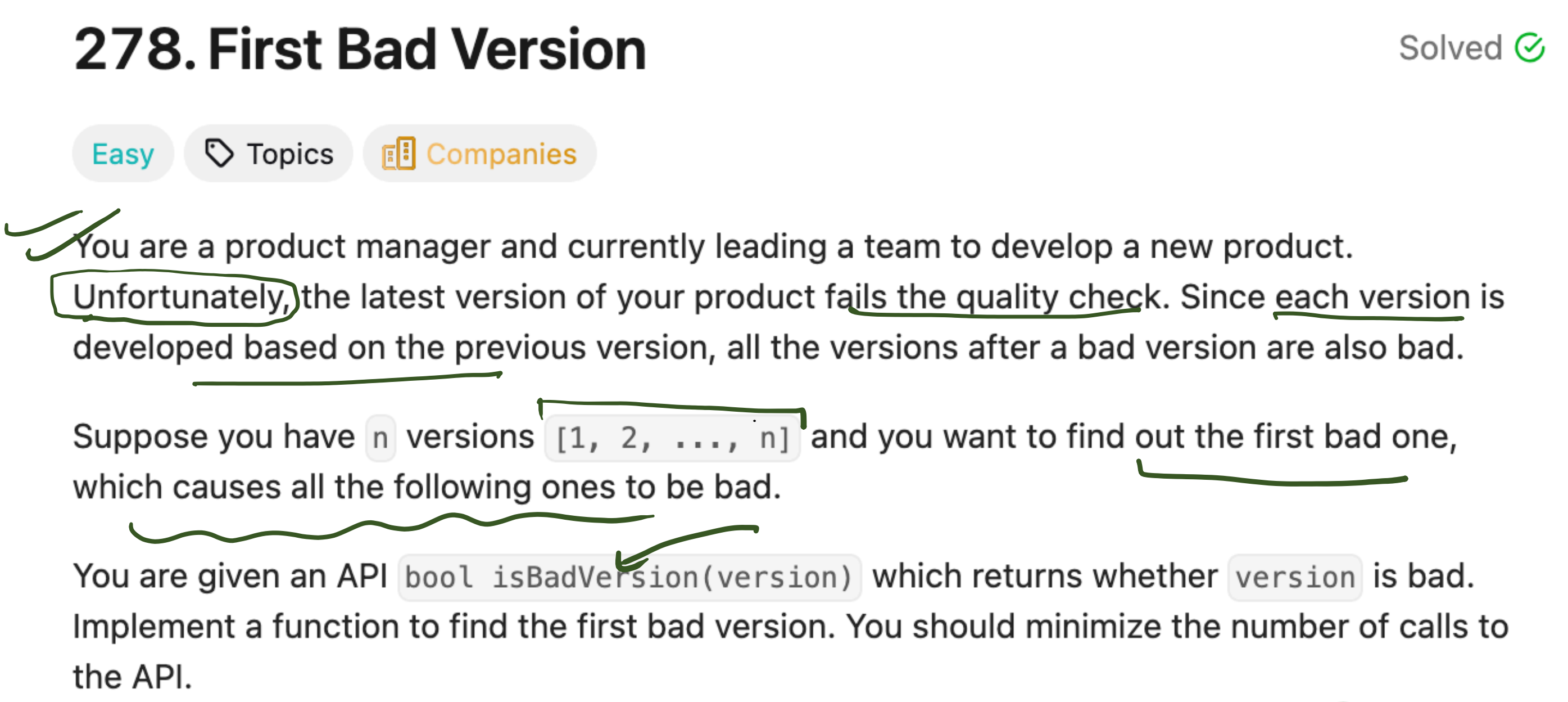
```
int x = 1
while (x^K <= n){
    x++
}
x-1
```



lo = 1
hi = n

```
while (lo <= hi){
    mid = (lo+hi)/2
    if(mid^K <= n){
        ans = mid
        lo = mid+1
    }
    else
        hi = mid-1
}
```

$75^3 \le 149$
$37^3 \le 14$
$18^3 \le$
$9^3 \le 14$
$4^3 \le 149$
$5^3 \le 1$



278. First Bad Version

Solved ✓

Easy Topics Companies

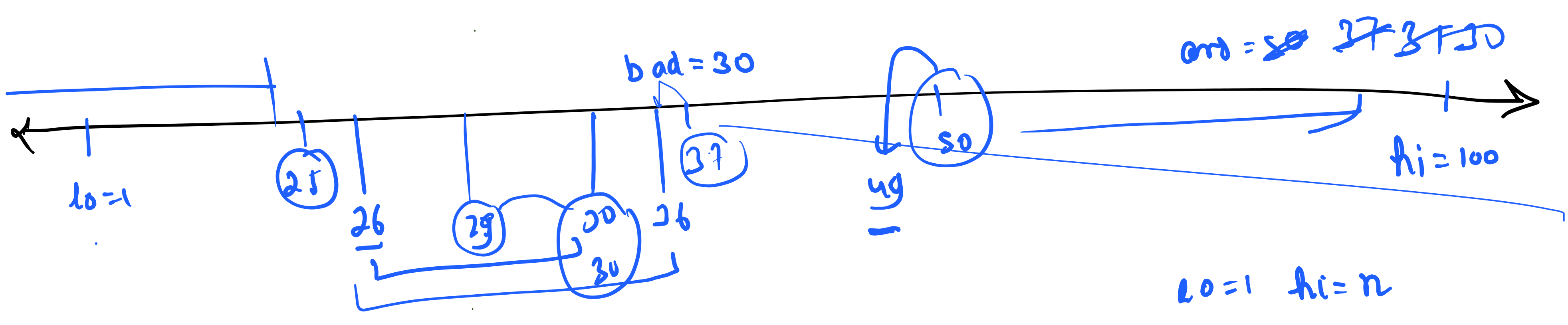
You are a product manager and currently leading a team to develop a new product. Unfortunately, the latest version of your product fails the quality check. Since each version is developed based on the previous version, all the versions after a bad version are also bad.

Suppose you have `n` versions `[1, 2, ..., n]` and you want to find out the first bad one, which causes all the following ones to be bad.

You are given an API `bool isBadVersion(version)` which returns whether `version` is bad. Implement a function to find the first bad version. You should minimize the number of calls to the API.

N = 100

bad = 30



```
lo = 1  hi = n
while(lo <= hi){
    mid = (lo+hi)/2
    if(isbadversion(mid)){
        ans = mid
        hi = mid-1
    }
    else
        lo = mid+1
}
```

$N = 2^{31} - 1$

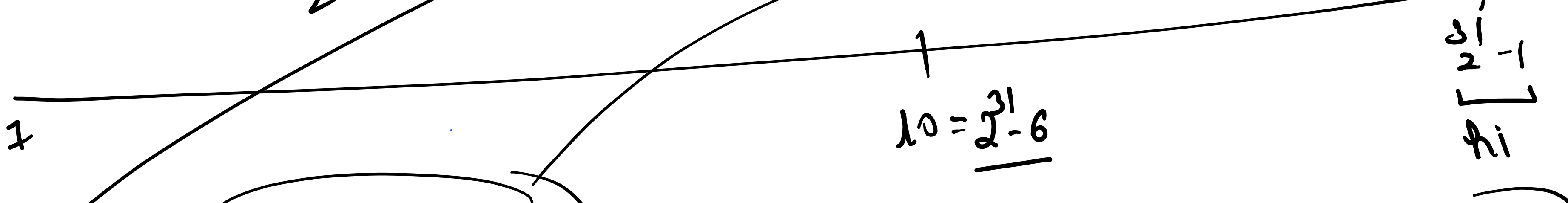
bad $= 2^{31} - 3$

$\frac{3+7}{2} = 5$

$3 + \left(\frac{7-3}{2}\right) = ⑤$

$\frac{lo+hi}{2} \rightarrow \frac{lo}{2} + \frac{hi}{2} = 1 + 3 = ④$

$\frac{3}{2}, \frac{7}{2}$

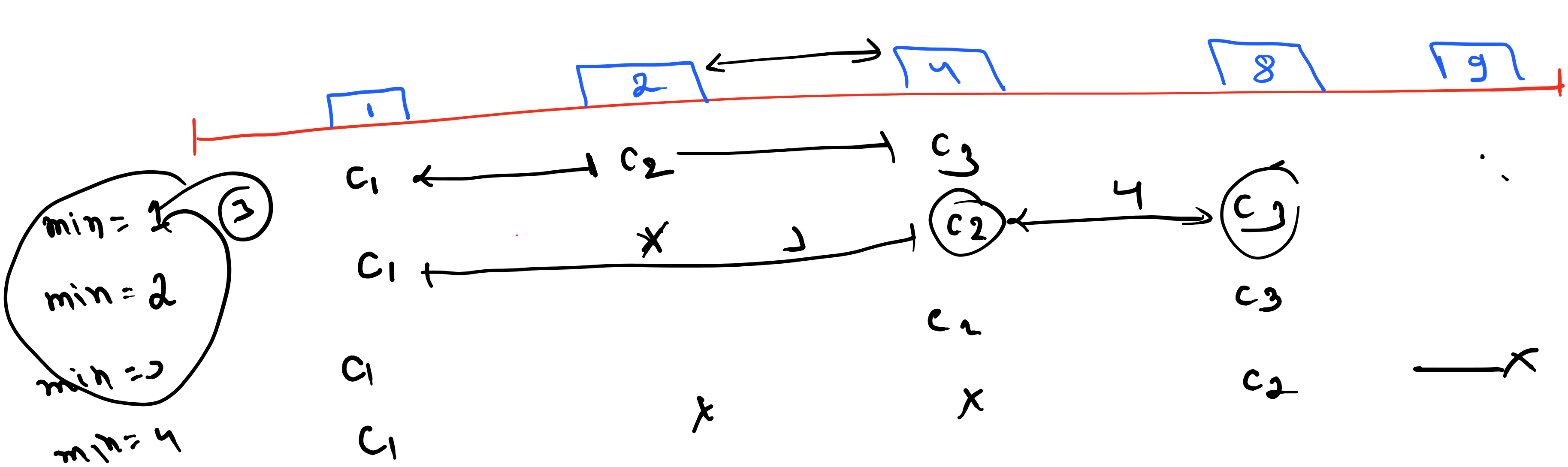


$$mid = \boxed{lo + \frac{(hi - lo)}{2}} = \frac{2 \times lo + hi - lo}{2} = \frac{lo + hi}{2}$$

$$\frac{lo+hi}{2} = \frac{2^{31} - 6 + 2^{31} - 1}{2} = \frac{2 \times 2^{31} - 7}{2} = \frac{\boxed{2^{32} - 7}}{2}$$

Chance

(1, 2, 4, 8, 9) ← [1 2 8 4 9] cow = 3 ✓



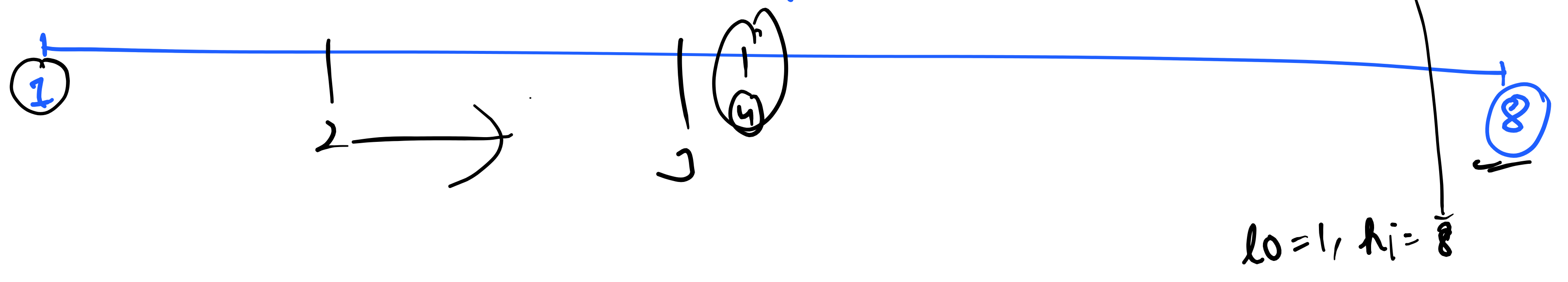
dis → 1
→ 2
→ 3 ✓
①

⑥ ← isPossible(Stall, noc, dis) → True
↳ Fa

[1, 2, 4, 8, 9]
0 1 2 3 4

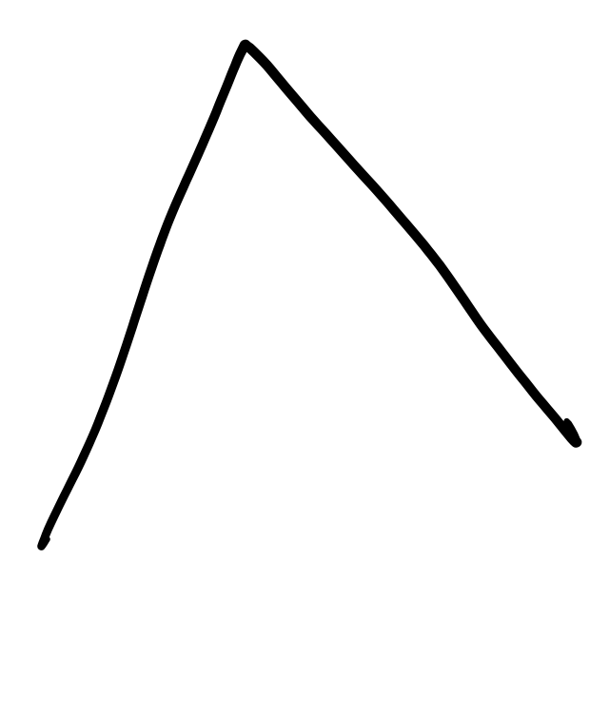
x - y = Gap

Stall[n-1] - Stall(0)



lo = 1, hi = 8

```
while (lo <= hi){
    mid = (lo+hi)/2
    if(isPossible(stall, noc, mid)){
        ans = mid
        lo = mid+1
    }
    else
        hi = mid
}
```



(1, 2, 4, 8, 9)

mid = 4

cow = ~~1~~ 2

pos = Stall(0) 4 ⑧

4-1 >= 3

2-1 >= 3

```
for(i=1   i<n   i++){
    if ( Stall(i) - pos >= mid){
        cow++
        pos = Stall(i)
    }
    if (cow == noc){
        return true
    }
}
false
```